MAT 344 HW04

Aditya Pahuja

Due 02/24

Problem 1. Let $f\colon \mathbb{C} \to \mathbb{C}$ be holomorphic. Find a weaker condition than being bounded above on $\mathbb{C}$ which implies that f is constant, and give the proof.

Solution. We will show that if $f \in \mathcal{O}(C)$ and there is a polynomial $p(z)$ of degree $N > 0$ such that

$$\lim_{|z|\to\infty} \frac{f(z)}{p(z)} = 0,$$

then f is a polynomial of degree at most $N-1$. In particular, the $N = 1$ case, in which $\frac{f(z)}{z}$ has limit zero, is one criterion for being constant.

Let f be as in the claim and suppose that $a_0, a_1, \cdots \in \mathbb{C}$ are such that $f(z) = \sum_{n=0}^{\infty} a_n z^n$ in a neighborhood of 0. The limit condition is equivalent to

$$\lim_{|z|\to\infty} \frac{f(z)}{p(z)} \cdot \frac{p(z)}{z^N} = 0$$

since the limit of $\frac{p(z)}{z^N}$ is the leading coefficient of p. Define $q(z) = a_0 + a_1 z + \cdots + a_{N-1} z^{N-1}$ and

$$g(z) = \frac{f(z) - q(z)}{z^N} \in \mathcal{O}(\mathbb{C} - \{0\}).$$

Since the degree of q is smaller than N, $g(z)$ also has limit zero as $|z|$ grows arbitrarily large, and g extends to a holomorphic function $\tilde{g} \in \mathcal{O}(\mathbb{C})$ by setting $\tilde{g}(0) = a_N$, since $\tilde{g}(z) = a_N z^N + a_{N+1} z^{N+1} + \cdots$ in a neighborhood of 0. The limit condition implies $\tilde{g}$ is bounded, hence constant by Liouville's theorem, so

$$\frac{f(z) - q(z)}{z^N} = \tilde{g}(z) = \tilde{g}(0) = a_N$$

for all z, implying that $f(z) = a_0 + a_1 z + \cdots + a_{N-1} z^{N-1} + a_N z^N$. However, the only way that this is compatible with the limit of $\frac{f(z)}{z^N}$ being zero is if $a_N = 0$, proving the claim. $\square$

Problem 2. Let $\gamma\colon S^1 \to \mathbb{C}$ be a regular curve, possibly with self-intersections. Show that for $a \notin \operatorname{im} \gamma$,

$$\frac{1}{2\pi i} \int_\gamma \frac{\mathrm{d}z}{z - a}$$

is the number of times that γ goes around a.

Solution. Suppose without loss of generality that $a = 0$. Let $\omega\colon S^1 \equiv \mathbb{R}/2\pi\mathbb{Z} \to \mathbb{C}$ be defined by $\omega(t) = \gamma(0) \cdot e^{i\theta}$. Recall that ω generates the fundamental group $\pi_1(\mathbb{C} - \{0\}, \gamma(0))$, so there exists an integer d such that γ is path homotopic to ω concatenated with itself d times (or the reverse path $t \mapsto \omega(2\pi - t)$ is concatenated with itself $|d|$ times, if $d < 0$); this integer d is what we will take to be the "number of times that γ goes around a," where the sign tells us the orientation of the curve γ. Since line integrals are invariant under path homotopy,

$$\frac{1}{2\pi i} \int_\gamma \frac{\mathrm{d}z}{z} = d \cdot \frac{1}{2\pi i} \int_\omega \frac{\mathrm{d}z}{z} = d$$

where the last equality is because ω is a parametrization of the boundary of an open ball centered at 0. $\square$

Problem 3. Let $U \subseteq \mathbb{C}$ be an open set. Suppose $f \in \mathcal{O}(U)$. Show that f cannot have a local maximum in U unless f is constant.

Solution. For any $z \in U$ and open ball $B \equiv B_r(z) \subseteq U$ centered at z with radius r, Cauchy's integral formula says

$$f(z) = \frac{1}{2\pi i} \int_{\partial B} \frac{f(\zeta)}{\zeta - z} \mathrm{d}\zeta = \frac{1}{2\pi} \int_0^{2\pi} f(z + re^{i\theta}) \mathrm{d}\theta.$$

Taking magnitudes,

$$|f(z)| \leq \frac{1}{2\pi} \int_0^{2\pi} \left| f(z + re^{i\theta}) \right| \mathrm{d}\theta \leq \sup_{\theta \in [0, 2\pi]} \left| f(z + re^{i\theta}) \right|$$

so there is a point on ∂B whose image under f is farther from 0 than $f(z)$. Since this argument works for any sufficiently small r, we can find points w arbitrarily close to z such that $|f(w)| > |f(z)|$, meaning z is not a local maximum for f. $\square$

Problem 4. Let Ω be a good domain, and suppose f is holomorphic on a neighborhood of $\overline{\Omega}$. Suppose $a \in \mathbb{C}$ satisfies $a \notin f(\partial\Omega)$.

(i) Show that the number $N = N(a)$ of times a is taken in Ω (counting multiplicity) is given by

$$N = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f'(\zeta)}{f(\zeta) - a} \mathrm{d}\zeta.$$

(ii) Let $z_1, \ldots, z_N$ be the points in Ω, counting multiplicity, where $f = a$. Show that if g is holomorphic in a neighborhood of $\overline{\Omega}$, then

$$\sum_{k=1}^{N} g(z_k) = \frac{1}{2\pi i} \int_{\partial\Omega} g(\zeta) \frac{f'(\zeta)}{f(\zeta) - a} \mathrm{d}\zeta.$$

Solution. We prove (ii), and then (i) follows from the special case $g \equiv 1$. For each z_k, let B_k be an open ball centered at z_k which is contained in Ω and on which f has a power series expansion centered at z_k. Consider $\Lambda = \Omega - \bigcup_{k=1}^{N} B_k$: since $f(\zeta) - a$ is nonzero on Λ, Green's theorem implies that

$$\frac{1}{2\pi i} \int_{\partial\Lambda} g(\zeta) \frac{f'(\zeta)}{f(\zeta) - a} \mathrm{d}\zeta = 0,$$

so

$$\frac{1}{2\pi i} \int_{\partial\Omega} g(\zeta) \frac{f'(\zeta)}{f(\zeta) - a} \mathrm{d}\zeta = \frac{1}{2\pi i} \sum_{k=1}^{N} \int_{\partial B_k} g(\zeta) \frac{f'(\zeta)}{f(\zeta) - a} \mathrm{d}\zeta.$$

For a given $k \in \{1, \ldots, N\}$ let d_k be the unique integer such that $f(z) - a = (z - z_k)^{d_k} h(z)$ for a function $h \in \mathcal{O}(B_k)$ with $h(z_k) \neq 0$. Then

$$\frac{f'(z)}{f(z) - a} = \frac{d_k (z - z_k)^{d_k - 1} h(z) + (z - z_k)^{d_k} h'(z)}{(z - z_k)^{d_k} h(z)} = \frac{d_k}{z - z_k} + \frac{h'(z)}{h(z)}.$$

In the integral,

$$\int_{\partial B_k} g(\zeta) \frac{f'(\zeta)}{f(\zeta)} \mathrm{d}\zeta = \int_{\partial B_k} \frac{d_k g(\zeta)}{\zeta - z_k} + \frac{g(\zeta) h'(\zeta)}{h(\zeta)} \mathrm{d}\zeta = 2\pi i \cdot d_k g(z_k)$$

where, in the second integral, the first addend of the integrand integrates to $2\pi i \cdot d_k g(z_k)$ by Cauchy's integral formula and the second addend integrates to zero because it is holomorphic on B_k. Summing over all k and dividing $2\pi i$ gives the desired result. $\square$